

AI Tools for Data Science and Statistics

140.697.79, Summer Institute, 2026–2027

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Instructor

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Course Description

AI tools can accelerate applied work in data science and statistics, but many analysts use them in ad-hoc ways that are inefficient, unreproducible, or unsafe. This course teaches structured, repeatable AI-assisted workflows for coding, data cleaning, debugging, documentation, simulation, and pipeline development. Topics include chatbot and harness selection, context management, skill files, agent workflows, project organization, privacy-aware sandboxing, and cost control. Exercises use R for illustration, but the concepts generalize to any statistical language.

Course Details

Course Number	140.697.79
Term	Summer Institute
Academic Year	2026–2027
Credits	1
Dates	June 15, June 16, and June 17, 2026
Time	9:00–10:50 AM ET
Location	Internet
Instruction Method	Online synchronous
Resources	CoursePlus
Grading	Letter Grade or Pass/Fail
Auditors	Allowed with instructor consent
Undergraduates	Allowed
Prerequisites	No prerequisites
Enrollment	Unrestricted
Instructor Consent	No consent required
Frequency Schedule	One Year Only; only offered in 2026
Format	Lecture, live demos, discussion, hands-on exercises

Course Learning Objectives

By the end of this course, students will be able to:

- Design and implement structured AI-assisted workflows to support coding, data cleaning, debugging, simulation, and documentation.
- Use LLMs and agent-based tools effectively within statistical workflows.
- Critically evaluate AI-generated code, outputs, and recommendations for accuracy, appropriateness, and alignment with analytic goals.
- Apply principles of responsible AI use, including privacy protection, data safety, and ethical considerations when working with sensitive or regulated data.
- Create and manage context files, skill files, and agent configurations to improve model performance.
- Identify and mitigate common pitfalls in AI-assisted statistical computing, such as model inaccuracy, hallucinations, and output bloat.
- Integrate AI tools into epidemiologic and public health data analysis pipelines to improve efficiency, transparency, and reproducibility.

Schedule

Session 1 (Monday, June 15): Introduction, Agents, and Context

We will introduce the course workflow and study the practical landscape of AI tools. The first part of class will provide a condensed introduction to LLMs, why they are useful, where they fail, and why structured workflows matter.

- **Problem set 1 assigned:** AI-assisted data visualization with cleaning, critique, and reflection.

Session 2 (Tuesday, June 16): Models, Agents, and Structured Workflows

We will focus on model selection, agents, and repeatable AI-assisted data science workflows. We will discuss cost, speed, reasoning behavior, coding ability, cloud/local tradeoffs, and when to use different models.

- **Problem set 2 assigned:** AI-assisted statistical modeling with diagnostics and verification.

Session 3 (Wednesday, June 17): Privacy And Simulated Data Workflows

We will study and practice privacy-aware AI-assisted workflows. The class will focus on simulated-data workflows for sensitive or regulated analysis.

- **Final project:** We will start the final project in class; the final project is due July 3, 2026.

Pre-Work

Before the first class, students should complete the [Pre-Class Setup Checklist](#). In brief, students should:

- Install Git or confirm that Git is available on their computer.
- Install R, RStudio or Positron, and Quarto. You may use alternative tools if you are comfortable with them.
- Arrange access to at least one AI tool or agent workflow for the course.
- Clone the course materials and use `git pull` for updates.

A GitHub account is only needed if a student's chosen AI tool requires it, such as GitHub Copilot. Personal access tokens are not part of the course workflow.

Course Project

Each student will complete a simulated-data privacy workflow. The goal is to choose a real dataset, create a simulated version for AI-assisted development, verify the code, then run the reviewed code on the real dataset outside the AI tool. Students may use their own non-sensitive data after discussing it in class. See the [final project page](#) for full details.

Date	Milestone
June 15 (Session 1)	Problem Set 1 assigned
June 16 (Session 2)	Problem Set 2 assigned; choose final project real dataset and simulation plan
June 17 (Session 3)	Final project introduced and started in class
July 3	Final project due via CoursePlus

Tools and Subscriptions

Students should arrange access to at least one AI tool or agent workflow for the course. A one-month paid subscription is often useful and typically costs about \$20, but existing access or free trials are fine. Options include:

Tool	Cost
OpenAI ChatGPT Plus	\$20/month
Anthropic Claude Pro	\$20/month
Google Gemini Advanced	\$20/month (one month free trial)
Cursor Pro	\$20/month
GitHub Copilot Pro	\$10/month

Variety across the class is encouraged; we will compare how different tools handle the same problems. If financing is a concern, please reach out to the instructor.

Students should also ensure that Git is installed before the first class. Mac and Linux typically have Git pre-installed. Otherwise, follow directions at git-scm.com.

Course Materials

There is no required textbook. Course readings and further reading are listed on the [course website](#). Course materials will be distributed through CoursePlus and the public course repository.

Methods of Assessment

This course is evaluated as follows:

- **Participation (30%):** Attend all sessions and engage actively in discussions and hands-on exercises.
- **Problem sets (30%):** Complete applied visualization and modeling exercises that practice structured AI-assisted workflows.
- **Final project (40%):** Complete a simulated-data privacy workflow at home with separate **simulated/** and **real/** folders and submit a reflection.

Generative AI Policy

Using AI tools is the subject of this course. Their use is permitted, encouraged, and expected. Students are responsible for understanding, verifying, and explaining any AI-assisted work they submit.

Academic Ethics

Students enrolled in the Bloomberg School of Public Health of The Johns Hopkins University assume an obligation to conduct themselves in a manner appropriate to the University's mission as an institution of higher education. Students should be familiar with the policies and procedures specified under Policy and Procedure Manual Student-01 (Academic Ethics) and the Student Conduct Code (Student-06), available at my.publichealth.jhu.edu.

Student Health and Well-being

If you are struggling with anxiety, stress, depression, or other mental health related concerns, please consider connecting with resources:

- Student support: bit.ly/bsphstudentsupport
- Mental Health Services: wellbeing.jhu.edu/MentalHealthServices
- Behavioral Health Crisis Support Team (24/7): 410-516-9355

Disability Accommodations

Student Disability Services (SDS) provides accessible and inclusive educational experiences for students with disabilities. To request accommodations:

1. Complete the SDS online application via [AIM](#)
2. Submit documentation using the provided link after application submission
3. Schedule a meeting with [Audrey Ndaba](#)

More information: [Student Disability Services](#)